TMA4180 Optimization 1

Sarah Pfalzer, Sunniva Kinnerud

March 2026

Introduction

In modern day infrastructure planning, the optimal placement of certain buildings plays a central role. If we were to decide the placement of a new hospital, we would want to optimise its location such that as many people as possible have the shortest distance to travel in case of an emergency. This is an example of a location problem; we wish to find one or more locations with minimal distance to a set of known locations. The goal of this report is to build a theoretical foundation to solve different location problems using different distance measurements. We then look at concrete algorithms that approximate the solution of a more complicated location problem.

Theoretical Foundation

When a location problem is to be modelled, distance measurements and the selected objective functions play an important role. In this report we consider the following distance measurements:

- $d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|$ (Manhattan or Taxicab distance)
- $d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$ (Euclidean distance)
- $d_\infty(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$ (Maximum distance)

We also consider m different points in the plane, $a^i = (a_1^i, a_2^i)^T \in \mathbb{R}^2, \quad i = 1, ..., m$. We write $A = \{a^1, ..., a^m\}$ as the set of all given locations in the plane. Thus the distance between the placement of a new hospital, given by x, and different neighbouring facilities given by a^i can vary depending on the distance measurement chosen. The choice of the distance measurement can vary with application; if we want linear distance we might choose the Euclidean distance measurement, or if movement in both directions is possible, but only the larger of the two determines the length of the movement we might choose the Maximum distance [1]. In addition to the different distance measurements, we will also consider two different objective functions:

- $\max_{a \in A} d(a, x)$ (Center problem)
- $\sum_{a \in A} d(a, x)$ (Median function)

The location problems can thus be reformulated to the following minimisation problems:

$$\min_{x \in \mathbb{R}^2} \max_{a \in A} d(a, x) \tag{1}$$

and

$$\min_{x \in \mathbb{R}^2} \sum_{a \in A} d(a, x) \tag{2}$$

where (1) is called the Center problem, and (2) is the Median problem, also called the Fermat-Weber problem.
In our location problems, the distance measurements directly influence the objective functions. If they are metrics, it will ensure that the distances are consistent and well-behaved. We therefore first start by showing that our three distance measurements are metrics.

To show that d_1, d_2 and d_∞ are metrics, the following three conditions must hold for all $x, y, z \in Y$, where Y is a non-empty set of $\mathbb{R}^2$:

- $d(x, y) = 0 \Leftrightarrow x = y$ *(definiteness)*
- $d(x, y) = d(y, x)$ *(symmetry)*
- $d(x, z) \leq d(x, y) + d(y, z)$ *(triangle inequality)*

The metric d_1 is defined as $d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|$. As both the terms are positive, $d_1(x, y)$ can only be 0 if $x_1 = y_1$ and $x_2 = y_2$. This holds the other way too: if $x_1 = y_1$ and $x_2 = y_2$, $d_1(x, y) = 0$, proving the first condition.

As the two terms are absolute values, and we consider the coordinates of x and y, which are scalars, $|x - y| = |y - x|$, and thus the symmetry condition is satisfied.

For the triangle inequality, we have $d_1(x, z) = |x_1 - z_1| + |x_2 - z_2| = |(x_1 - y_1) + (y_1 - z_1)| + |(x_2 - y_2) + (y_2 - z_2)|$. Using the triangle inequality for the absolute values gives $|(x_1 - y_1) + (y_1 - z_1)| \leq |x_1 - y_1| + |x_1 - z_1|$, and the same for the second term. Thus $d_1(x, z) \leq |x_1 - y_1| + |y_1 - z_1| + |x_2 - y_2| + |y_2 - z_2| = d_1(x, y) + d_1(y, z)$, proving the triangle inequality. $d_1(x, y)$ satisfies all three conditions, which proves that it is a metric.

The metric d_2 is defined by $d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$. As the square is always bigger than or equal to zero, the sum $(x_1 - y_1)^2 + (x_2 - y_2)^2$ can never be zero unless both the terms are zero. This only happens when $x = y$. This holds the other way too, and the first condition is satisfied.

As the subtraction is squared, $(x-y)^2 = (y-x)^2$, and d_2 satisfies the second condition.

For the triangle inequality, we recognize that this metric is induced by the Euclidean norm, where the triangle inequality holds. Thus $d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} = ||(x_1 - y_1, x_2 - y_2)||_2 = ||(x_1, y_1) - (x_2, y_2)||_2 = ||(x_1, x_2) - (z_1, z_2) + (z_1, z_2) - (y_1, y_2)||_2 = ||(x_1 - z_1, x_2 - z_2) + (z_1 - y_1, z_2 - y_2)||_2 \leq ||(x_1 - z_1, x_2 - z_2)||_2 + ||(z_1 - y_1, z_2 - y_2)||_2 = \sqrt{(x_1 - z_1)^2 + (x_2 - z_2)} + \sqrt{(z_1 - y_1)^2 + (z_2 - y_2)^2} = d_2(x, z) + d_2(z, y)$, proving the triangle inequality. $d_2(x, y)$ satisfies all three conditions, and is thus a metric.

The metric d_∞ is defined as $d_\infty = \max\{|x_1 - y_1|, |x_2 - y_2|\}$. This metric can only be zero if both terms are zero, because of the absolute value. Thus $d_\infty(x, y)$ can only be zero if $x = y$, and the same holds in the other direction.

Symmetry follows from the absolute values of the scalars in the max norm.

To show the triangle inequality, we first rewrite $d_\infty(x, y) = \max\{|x_1 - z_1|, |x_2 - z_2|\} \leq \max\{|x_1 - y_1| + |y_1 - z_1|, |x_2 - y_2| + |y_2 - z_2|\}$. Let $M_1 = \max\{|x_1 - y_1|, |x_2 - y_2|\}$ and $M_2 = \max\{|y_1 - z_1|, |y_2 - z_2|\}$. As $|x_1 - y_1|$ is in M_1 and $|y_1 - z_1|$ is in M_2, the sum will satisfy the inequality $|x_1 - y_1| + |y_1 - z_1| \leq M_1 + M_2$. The same argument holds for the second terms, and thus $\max\{|x_1 - y_1| + |y_1 - z_1|, |x_2 - y_2| + |y_2 - z_2|\} \leq M_1 + M_2 = d_\infty(x, y) + d_\infty(y, z)$, which proves the triangle inequality. As d_∞ satisfies all three conditions, it is a metric.

Having established that all our chosen distance measurements are in fact metrics, we can now look at their geometric interpretation by plotting their respective unit balls. This will help us visualize how the choice of metric physically constrains the search space in an optimisation problem. In figure 1, the unit balls $B_i(0, 1) := \{\boldsymbol{x} \in \mathbb{R}^2 : ||\boldsymbol{x}||_i \leq 1\}, i = 1, 2, \infty$ are plotted.

The unit ball of the maximum norm, $B_\infty(0, 1)$, fills out the whole unit square $[0, 1] \times [0, 1]$. This is because the maximum norm for a vector is $||\boldsymbol{x}||_\infty = \max_i |x_i|$, and thus for all $\boldsymbol{x}$ where $x_1, x_2 \in [-1, 1]$ the maximum norm $||\boldsymbol{x}||_\infty \leq 1$.

The unit ball of the Euclidean norm, $B_2(0, 1)$, fills out the circle of radius 1 with centre in origo. The Euclidean norm for a (2×2)-vector is $||\boldsymbol{x}||_2 = \sqrt{x_1^2 + x_2^2}$, which is the formula for a circle with centre in origo and radius $||\boldsymbol{x}||_2$.

The unit ball of the Manhattan norm, $B_1(0, 1)$, fills out a diamond with corners in $(-1, 0), (0, -1), (1, 0)$ and $(0, 1)$. The Manhattan norm for a (2×2)-vector is $||\boldsymbol{x}||_1 = |x_1| + |x_2|$, i.e. the sum of the absolute values of the two vector-components. In the area not encompassed by the unit ball of the Manhattan norm, the sum of the x- and y-components surpass 1.

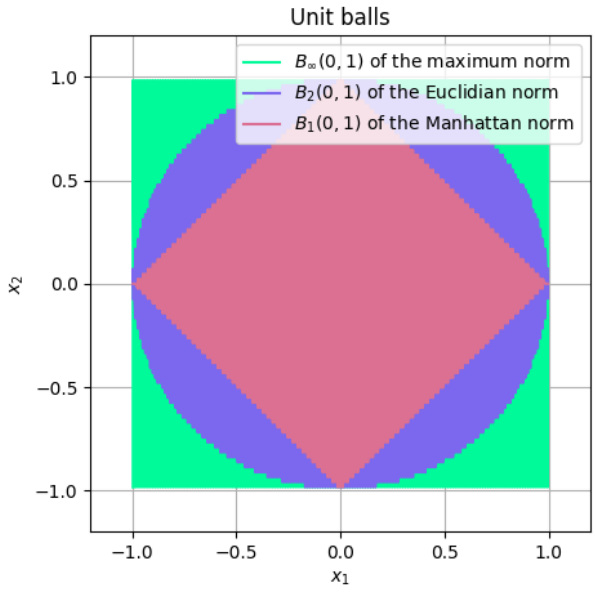


Figure 1: The figure shows three unit balls, of respectively the maximum norm, Euclidean norm and the Manhattan norm.

From Figure 1 we can also observe that each of the plotted regions appear as convex sets. This implies that the distance functions are induced by norms, whose unit balls are convex. Being convex is one of the most desirable properties of an objective function, as it ensures that any local minimum found by an algorithm is also a global minimum [2]. We

therefore wish to show that each of the objective functions is in fact convex, ensuring that any local minima is also a global minima. We do this by first showing that every norm is convex. Then, by assuming that our three given metrics are induced by norms, we can then show that our objective functions, the Center problem and the Median function are convex as well.
A function $||\cdot|| : \mathbb{R}^2 \to \mathbb{R}$ is called a norm on $\mathbb{R}^2$ if it fulfils the following conditions $\forall x, y \in \mathbb{R}^2$, and $\forall \alpha \in \mathbb{R}$:

- $||x|| = 0 \Leftrightarrow x = 0$ (definiteness)
- $||\alpha x|| = |\alpha| \cdot ||x||$ (positive homogeneity)
- $||x + y|| \leq ||x|| + ||y||$ (triangle inequality).

A function $f : \mathbb{R}^d \to \mathbb{R}$ is convex if $\forall x, y \in \mathbb{R}^d$ and $0 \leq \lambda \leq 1$, then $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$.

Thus, a norm $||\cdot|| : \mathbb{R}^2 \to \mathbb{R}$ is convex if $\forall x, y \in \mathbb{R}^2$ and $0 \leq \lambda \leq 1$, $||\lambda x + (1 - \lambda)y|| \leq \lambda||x|| + (1 - \lambda)||y||$. By the triangle inequality and positive homogeneity,

$$||\lambda x + (1 - \lambda)y|| \leq ||\lambda x|| + ||(1 - \lambda)y|| = |\lambda| \cdot ||x|| + |1 - \lambda| \cdot ||y|| = \lambda||x|| + (1 - \lambda)||y||,$$

which shows that every norm is a convex function.

To prove that the objective functions (1) and (2) are convex, we first assume that the given metrics are induced by norms, so $d(x, y) = ||x - y||$.

We first start by showing that $f(x) = ||x - a||$ is convex. Let $x, y \in \mathbb{R}^2$, and let $\lambda \in [0, 1]$. Then $||\lambda x + (1-\lambda)y - a||$. This can be rewritten as $||\lambda(x-a) + (1-\lambda)(y-a)|| \leq ||\lambda(x - a)|| + ||(1-\lambda)(y-a)|| = \lambda||x - a|| + (1-\lambda)||y - a||$ by the triangle inequality and positive homogeneity of norms. The function is thus convex.

The median objective function is given by

$$g(x) = \sum_{i=1}^{m} f_i(x),$$

where $f_i(x) = ||x - a_i||$. This function is convex by $g(\lambda x + (1-\lambda)y) = \sum_i f_i(\lambda x + (1-\lambda)y) \leq \sum_i(\lambda f_i(x) + (1 - \lambda)f_i(y)) = \lambda \sum_i f_i(x) + (1 - \lambda) \sum_i f_i(y) = \lambda g(x) + (1 - \lambda)g(y)$.

The Center problem is given by

$$g(x) = \max_{i=1,..m} f_i(x),$$

where $f_i(x) = ||x - a_i||$. From the lecture notes [2], we know that if two functions f_1 and f_2 are convex, then the function defined by $h_2 = \max\{f_1, f_2\}$ is convex. Then $h_3 = \max\{h_2, f_3\} = \{f_1, f_2, f_3\}$ is also convex, as h_2 is convex. Thus, $g(x) = \max\{f_1, ..., f_m\}$ is convex by induction. Because of this, the two objective functions are convex.

Analysis of Center and Median problems

Having established the convexity of the d_2 norm and the Center problem given in (1), we can give a concrete geometrical interpretation for solving the Center problem with the Euclidean distance metric. The Center problem in the assignment is defined as

$$\min_{x \in \mathbb{R}^2} \max_{a \in A} d(a, x),$$

where $d(a, x)$ is given by the Euclidean distance $d_2(a, x) = \sqrt{(x_1 - a_1)^2 + (x_2 - a_2)^2}$. We wish to choose a placement in the plane, given by x to minimise the maximum distance to all locations in the set A. Because the metric used is the Euclidean norm, this becomes a circle centred in x, as shown by the unit ball in Figure 1, with radius equal to the smallest distance that encloses all the points in A. This is illustrated in Figure 2.

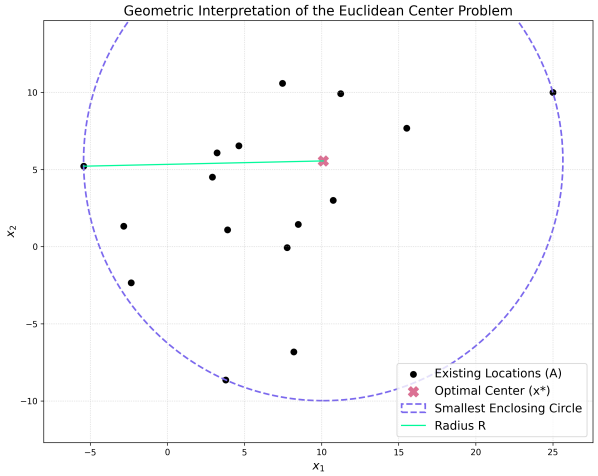


Figure 2: The figure shows the geometric interpretation of the Center problem with Euclidean distance.

Another problem that allows us to use previous results to give a solution approach is the Median problem (2) with the Manhattan distance function. Contrary to the last problem, this problem allows us to use separability to find an exact solution. As previously stated, the Median problem is defined as

$$\min_{x \in \mathbb{R}^2} \sum_{a \in A} d_1(a, x),$$

where $d_1(x, y) = |x_1 - a_1| + |x_2 - a_2|$. The objective function is thus $f(x) = \sum_{a \in A}(|x_1 - a_1| + |x_2 - a_2|)$. This can be separated into two objective functions $f_1(x_1) = \sum_{a \in A} |x_1 - a_1|$ and $f_2(x_2) = \sum_{a \in A} |x_2 - a_2|$, which can be solved separately for x_1 and x_2 respectively. For the objective function $f_1(x_1)$, the solution to the minimisation problem is a point along the x-axis that minimises the sum to all the x-coordinates of the points in the set A.

A way of solving this problem can be by first sorting the x-coordinates of the points in the set A. Let the starting point be the median of the coordinates. For an odd number of points in the set A, this is a unique point, for even number of points in the set A, this will be a point in between the middle two points. If we move a distance δ_r to the right of the median, the distance to all the points to the right will decrease by δ_r. Similarly, the distance to all the points to the left will increase by δ_r. As we are in the median, the net difference between the increase and decrease in distances will be zero. The only difference left is the distance to the median point itself, which has increased. The same argument can be used when moving to the left of the median point. Because the objective function is convex, all local minimisers are global minimisers. As the median is a local minimiser, it is thus a global minimiser. The same argument can be made for the second objective function $f_2(x_2)$. The solution to this minimisation problem is thus the componentwise median of the coordinates of the points in A.

Having examined the separability of the Manhattan norm, we return to Euclidean space. More precisely, we want to consider the Median problem (2) with a squared and a standard Euclidean norm. As we will see, the smooth nature of the squared Euclidean norm makes for an analytical solution. On the other hand, deriving necessary optimality conditions for the Euclidean norm proves a bit more difficult, as it is not differentiable. We first start by looking at the Median problem with the squared Euclidean norm. The median problem $\min_{x\in\mathbb{R}^2} \sum_{a\in A}(d_2(a,x))^2$ has objective function $f(x) = \sum_{a\in A}(d_2(a,x))^2$. This function can be shown to be convex using proposition 3.16 [2], which states that $f \in C^2$ is convex if and only if $H_f(x)$ is positive semidefinite $\forall x \in \mathbb{R}^d$. The proposition holds for f since it is a second-degree polynomial.

Writing out the objective function gives

$$f(x) = \sum_{a\in A}(d_2(a,x))^2 = \sum_{a\in A}((a_1 - x_1)^2 + (a_2 - x_2)^2) = \sum_{a\in A}(a_1 - x_1)^2 + \sum_{a\in A}(a_2 - x_2)^2,$$

which can be used to calculate the Hessian

$$H_f(x) = \nabla^2 f(x) = \begin{bmatrix} \partial^2_{x_1} f & \partial_{x_1x_2} f \\ \partial_{x_2x_1} f & \partial^2_{x_2} f \end{bmatrix} = \begin{bmatrix} 2m & 0 \\ 0 & 2m \end{bmatrix},$$

where m is the number of fixed points in the set A. As this is a diagonal matrix, the eigenvalues are the elements on the diagonal, i.e. $\lambda = 2m > 0$, which means that the Hessian is positive definite, and hence also positive semidefinite. By the proposition, the objective function is therefore convex. We note that since the Hessian is positive definite, by remark 3.17 [2] the objective function is also strictly convex.

The uniquely determined minimiser of this problem can be computed analytically using corollary 3.12 [2], which states that for a function $f : \mathbb{R}^d \to \mathbb{R}$ that is convex and differentiable, x^* is a global minimiser if and only if $\nabla f(x^*) = 0$. The corollary holds because f is a second-degree polynomial. Setting the gradient of f to zero gives

$$\nabla f(x) = \begin{bmatrix} \partial_{x_1} f \\ \partial_{x_2} f \end{bmatrix} = \begin{bmatrix} -2\sum_{a\in A}(a_1 - x_1) \\ -2\sum_{a\in A}(a_2 - x_2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which in turn gives

$$\begin{aligned} \sum_{a\in A} a_1 = \sum_{a\in A} x_1 \text{ and } \sum_{a\in A} a_2 = \sum_{a\in A} x_2 \\ \implies x_1 = \frac{1}{m}\sum_{a\in A} a_1 \text{ and } x_2 = \frac{1}{m}\sum_{a\in A} a_2. \end{aligned}$$

The minimiser of this median problem is $x^* = (\frac{1}{m}\sum_{a\in A} a_1, \frac{1}{m}\sum_{a\in A} a_2)^T$. We note that this vector has the mean of the given locations in x-direction as the x-coordinate, and the mean of the given locations in y-direction as the y-coordinate.

We now want to minimise the objective function with the non-squared Euclidean norm, given by

$$\sum_{a\in A}\sqrt{(a_1 - x_1)^2 + (a_2 - x_2)^2}.$$

As discussed previously, we know that the objective function is convex. However, it is not continuously differentiable when $x = a$. In this case the necessary optimality conditions are defined by the subgradient of the objective function, given by $\partial f(x)$, and not just the gradient itself, as it were previously when we considered the squared Euclidean norm. A vector g is a subgradient of f at x if

$$f(z) \geq f(x) + g^T(z - x) \quad \forall z.$$

For continuously differentiable points, when $x \neq a$, the subgradient is just the gradient:

$$\partial f(x) = \left\{ \sum_{a \in A} \frac{x - a}{||x - a||_2} \right\} \qquad \forall a \in A.$$

When $x = a_k$, the subgradient becomes

$$||z - a_k||_2 \geq ||a_k - a_k||_2 + g^T(z - a_k) = g^T(z - a_k) \quad \forall z.$$

Let $y = z - a_k$. The condition then becomes

$$g^T y \leq ||y||_2.$$

Using Cauchy-Schwarz, we get

$$g^T y \leq ||g||_2 ||y||_2.$$

So if $||g||_2 \leq 1$, then $g^T y \leq ||g||_2 ||y||_2$, for all y, and $\{g : ||g|| \leq 1\} \subseteq \partial ||x - a_k||$ when $x = a_k$. If $g \in \partial ||x - a_k||_2 \Big|_{x=a_k}$, then

$$g^T y \leq ||y||_2 \quad \forall y.$$

Let $y = g$. Then $g^T g \leq ||g||_2$. We know that $g^T g = ||g||_2^2$, so $||g||_2^2 \leq ||g||_2$. Dividing by $||g||_2$, we get $||g||_2 \leq 1$, and thus $\partial ||x - a_k|| \Big|_{x=a_k} \subseteq \{g : ||g||_2 \leq 1\}$. If $||g||_2 = 0$, the inequality will hold as well. Thus

$$\partial ||x - a_k|| \Big|_{x=a_k} = \{g : ||g||_2 \leq 1\}.$$

Now that we know what the subgradient looks like, we can use it to derive the necessary optimality condition. For subgradients, we have the following statement:

Theorem 1. *x^* minimises $f \Leftrightarrow 0 \in \partial f(x^*)$*

Proof. If $0 \in \partial f(x^*) \Leftrightarrow f(z) \geq f(x^*) + 0^T(z - x^*) = f(x^*)\ \forall z$, so x^* minimises f. □

For points a_i, where $i \neq k$, the objective function is still differentiable, so we can use the gradient. Adding all the cases together using the sum rule for subgradients given by $\partial f(x) = \sum_{a \in A} \partial ||x - a||_2$, from [3], together with the case $i = k$ yields

$$\partial f(a_k) = \left\{ \sum_{i \neq k} \frac{a_k - a_i}{||a_k - a_i||_2} + g : ||g||_2 \leq 1 \right\},$$

as the subgradient of the objective function. To determine the necessary optimality condition, we use Theorem 1. From Theorem 1, we know that x^* minimises f if and only if $0 \in \partial f(x^*)$. Thus a_k minimises f if and only if

$$0 \in \left\{ \sum_{i \neq k} \frac{a_k - a_i}{||a_k - a_i||_2} + g : ||g||_2 \leq 1 \right\}.$$

This means that $0 = \sum_{i \neq k} \frac{a_k - a_i}{||a_k - a_i||_2} + g$ where $||g|| \leq 1$. Hence $\sum_{i \neq k} \frac{a_k - a_i}{||a_k - a_i||_2} = -g$ where $||g||_2 \leq 1$, so $|| \sum_{i \neq k} \frac{a_k - a_i}{||a_k - a_i||_2}||_2 = || - g|| = ||g|| \leq 1$. The optimality condition for the objective function at the discontinuous points are thus

$$|| \sum_{i \neq k} \frac{a_k - a_i}{||a_k - a_i||_2}||_2 \leq 1,$$

and the necessary optimality conditions are

$$\begin{cases} \dfrac{x - a_i}{\|x - a_i\|_2} = 0, & x^* \neq a_i \ \forall i, \\ \| \sum\limits_{i \neq k} \dfrac{a_i - a_k}{||a_i - a_k||_2}\|_2 \leq 1, & x^* = a_i. \end{cases}$$

Although the Euclidean norm is more difficult to minimise due to its non-differentiability, it might be the more optimal choice for a location problem, as we will now see. The global minimisers for the Median problem (2) with respectively Euclidean distance and squared Euclidean distance: $\min\limits_{x \in \mathbb{R}^2} \sum\limits_{a \in A} d_2(a, x)$ and $\min\limits_{x \in \mathbb{R}^2} \sum\limits_{a \in A} (d_2(a, x))^2$, do not coincide for all sets of given locations. An example of such a set is $A = \{(1, 0)^T, (2, 0)^T, (50, 0)^T\}$, where the minimiser for the Median problem with Euclidean distance function is $x^* = (2, 0)^T$ and the minimiser for the Median problem with squared Euclidean distance function is $x^* \approx (18, 0)^T$.

The problem $\min\limits_{x \in \mathbb{R}^2} \sum\limits_{a \in A} d_2(a, x)$ for the set A where the three points are in a line, reduces to a 1D problem. As the sum of the absolute values of distances are minimised by the median, this problem on the set A reduces to finding the median, which is 2. As for the other problem, where the distance function is squared, minimizing reduces to finding the average value. In this case, the average is approximately 18.

The average punishes big deviations in the set A noticeably more than the median, which means that often, sets with such "outliers" will result in different minimisers for the two problems. On the other hand, if the set is symmetric (e.g. four corners of a square), the minimisers will be the same. In an infrastructure context, using the squared Euclidean norm would pull the placement of a hospital significantly towards a distant village, which might increase travel time for the majority of the population. The non-squared Euclidean norm focuses on the central tendency to a higher degree. Thus, in this context, it might be a better measurement choice when we wish to decide the placement of a hospital.

The Weiszfeld Algorithm

As shown in the previous section, the Median problem (2) with the Euclidean distance proves more difficult to solve than the Median problem with the squared Euclidean norm.

This is mostly because we cannot solve for a minimiser directly, due to the distance measure not being differentiable in all points. We now turn to a similar problem; the Median problem with weighted Euclidean distances, given by

$$\min_{x\in\mathbb{R}^2} \sum_{i\in M} v^i d_2(a^i, x). \tag{3}$$

Here, v^i are the weights v^i, $i \in M = \{1, ..., m\}$, which are real non-negative numbers. As this objective function is the same as the Median problem with Euclidean norm multiplied by non-negative scalars, it is convex. Using this, we would be able to add weights to our points. These can for example represent population density when deciding the placement of a hospital in a populated area. This would result in a more realistic scenario, as a more densely populated area would hold a greater priority. As mentioned earlier, the minimiser for this type of problem is difficult to compute analytically. We therefore look at an iterative method to approximate the minimiser for this problem instead. This method is called the Weiszfeld algorithm.

The Weiszfeld algorithm consists of multiple steps. The first step consists of checking if one of the stationary points is itself a global minimiser. This check can be derived from our previously derived necessary optimality conditions for the Median problem with the Euclidean norm. As derived earlier, when the locale minimiser satisfy $x^* = a_i$, the necessary optimality condition is

$$\| \sum_{i\neq k} \frac{a_i - a_k}{||a_i - a_k||_2} \|_2 \leq 1,$$

which can be expanded to

$$\left(\left(\sum_{i\neq k} \frac{a_1^i - a_1^k}{d_2(a^k, a^i)} \right)^2 + \left(\sum_{i\neq k} \frac{a_2^i - a_2^k}{d_2(a^k, a^i)} \right)^2 \right)^{1/2} \leq 1.$$

Adding the weights we arrive at

Theorem 2. *If for*

$$Test_k := \left(\left(\sum_{i\in M\setminus\{k\}} \frac{a_1^k - a_1^i}{d_2(a^k, a^i)} \right)^2 + \left(\sum_{i\in M\setminus\{k\}} \frac{a_2^k - a_2^i}{d_2(a^k, a^i)} \right)^2 \right)^{1/2}$$

it holds that $Test_k \leq v^k$ for one $k \in M$, then $x^* = a^k$ is a global minimiser.

The first step in the algorithm thus consists of checking if this is true for any of the fixed points. If it is, we have found our minimiser, and the algorithm terminates.

Next we choose a starting point, (x_1, x_2). To get a starting point relatively close to our minimiser, we can choose this point to be the solution of the median problem using the squared Euclidean norm, as shown previously.

Next we derive the main iteration scheme for (x_1^{new}, x_2^{new}). As we are now looking at points that exclude the fixed points in the set A, we can look at the necessary optimality condition when $x \neq a_k$. As explained earlier, this is when $\nabla f(x) = 0$. For our objective function, given by

$$\sum_{i=1}^{m} v_i ||a_i - x||_2,$$

the optimality condition becomes

$$\sum_{i=1}^{m} v_i \frac{a_i - x}{||a_i - x||_2} = 0.$$

Splitting the sum yields

$$\sum_{i=1}^{m} \frac{v_i a_i}{||a_i - x||_2} - x \sum_{i=1}^{m} \frac{v_i}{||a_i - x||_2} = 0.$$

Solving this for x gives the main iteration scheme:

$$x = \frac{\sum_{i=1}^{m} v_i \frac{a_i}{||a_i - x||_2}}{\sum_{i=1}^{m} v_i \frac{1}{||a_i - x||_2}}.$$

This is a fixed point equation, which can be turned into an iteration. The main iteration scheme for the Weiszfeld algorithm is thus

$$x_1^{\text{new}} := \frac{\sum_{i=1}^{m} v^i \frac{a_1^i}{d_2(a^i,x)}}{\sum_{i=1}^{m} v^i \frac{1}{d_2(a^i,x)}},\ x_2^{\text{new}} := \frac{\sum_{i=1}^{m} v^i \frac{a_2^i}{d_2(a^i,x)}}{\sum_{i=1}^{m} v^i \frac{1}{d_2(a^i,x)}} \tag{4}$$

Our last step is to derive a stopping criterion for our algorithm. To do this, we first introduce a couple of theorems, which will later help us define the stopping criterion.

Theorem 3. *Let $x \in \mathbb{R}^2$ be an iteration point that was obtained during the Weiszfeld algorithm. Then*

$$UB(x) := ||\nabla f_{d_2}(x)|| \cdot \sigma(x)$$

is an upper bound for a possible improvement of the objective function $\nabla f_{d_2}(x)$, where $||\cdot||$ is the Euclidean norm in $\mathbb{R}^2$.

$\sigma(x)$ is defined as $\sigma(x) := \max\{d_2(x, y) : y \in \text{conv}\{a^1, ..., a^m\}\}$. The next theorem gives us an expression for the lower bound:

Theorem 4. *Let x^* denote the global minimiser of the problem* (3)*. A lower bound for $f_{d_2}(x^*)$ is given by*

$$LB(x) := f_{d_2}(x) - ||\nabla f_{d_2}(x)|| \cdot \sigma(x)$$

is an upper bound for a possible improvement of the objective function $\nabla f_{d_2}(x)$, where $||\cdot||$ is the Euclidean norm in $\mathbb{R}^2$.

From these two theorems, we can derive the relative accuracy of the current iterate of the Weiszfeld algorithm.

The relative accuracy of the current iterate x is defined as

$$|\frac{f_{d_2}(x) - f_{d_2}(x^*)}{f_{d_2}(x^*)}|[1].$$

From Theorem 2 we have that $f_{d_2}(x) - f_{d_2}(x^*) \leq ||\nabla f_{d_2}(x)||\sigma(x)$. From Theorem 3 we have $f_{d_2}(x) - ||\nabla f_{d_2}(x)|| \leq f_{d_2}(x^*)$. Dividing by this reverses the inequality sign, so

$$|\frac{f_{d_2}(x) - f_{d_2}(x^*)}{f_{d_2}(x^*)}| \leq |\frac{||\nabla f_{d_2}(x)||\sigma(x)}{f_{d_2}(x) - ||\nabla f_{d_2}(x)||\sigma(x)}| < \epsilon.$$

Thus, if we have a lower bound $LB(x) > 0$, and we have a given $\epsilon > 0$, if

$$\frac{UB(x)}{LB(x)} = \frac{||\nabla f_{d_2}(x)||\sigma(x)}{f_{d_2}(x) - ||\nabla f_{d_2}(x)||\sigma(x)} < \epsilon$$

is true, then the current iterate has at least a relative accuracy of ϵ.

From this, we can determine a stopping criterion for our algorithm. The previous result states that if $LB(x) > 0$ and for a given $\epsilon > 0$

$$\frac{UB(x)}{LB(x)} = \frac{||\nabla f_{d_2}(x)|| \cdot \sigma(x)}{f_{d_2}(x) - ||\nabla f_{d_2}(x)|| \cdot (x)} < \epsilon$$

holds true, then the current iterate of the Weiszfeld algorithm has a relative accuracy of a least ϵ. Using this theorem, we can derive a termination criterion for the algorithm: for a given $\epsilon > 0$, terminate the iteration scheme given by (4) once

$$\frac{UB(x^{\text{new}})}{LB(x^{\text{new}})} < \epsilon. \tag{5}$$

The implementation of the Weiszfeld algorithm described above can be found in the Jupyter Notebook attached. We used $\epsilon = 0.001$ for the termination criterion. For testing of the implemented algorithm, we have used four test cases with a set A of given locations and a set v of weights:

$$A_1 = \{(1,1),(-1,1),(1,-1),(-1,-1)\}, \quad v = \{1,1,1,1\} \tag{C1}$$

$$A_1 = \{(1,1),(-1,1),(1,-1),(-1,-1)\}, \quad v = \{1,1,1,2\} \tag{C2}$$

$$A_2 = \{(1,0),(2,0),(50,0)\}, \quad v = \{1,1,1\} \tag{C3}$$

$$A_3 = \{(-14,8),(-12,-4),(2,-12),(10,-5),(16,4),(5,15),(-2,5)\}, \quad v = \{5,3,7,9,2,5,1\} \tag{C4}$$

Figure 3 shows the given locations a^i from the set A along with the minimiser found with the Weiszfeld algorithm.

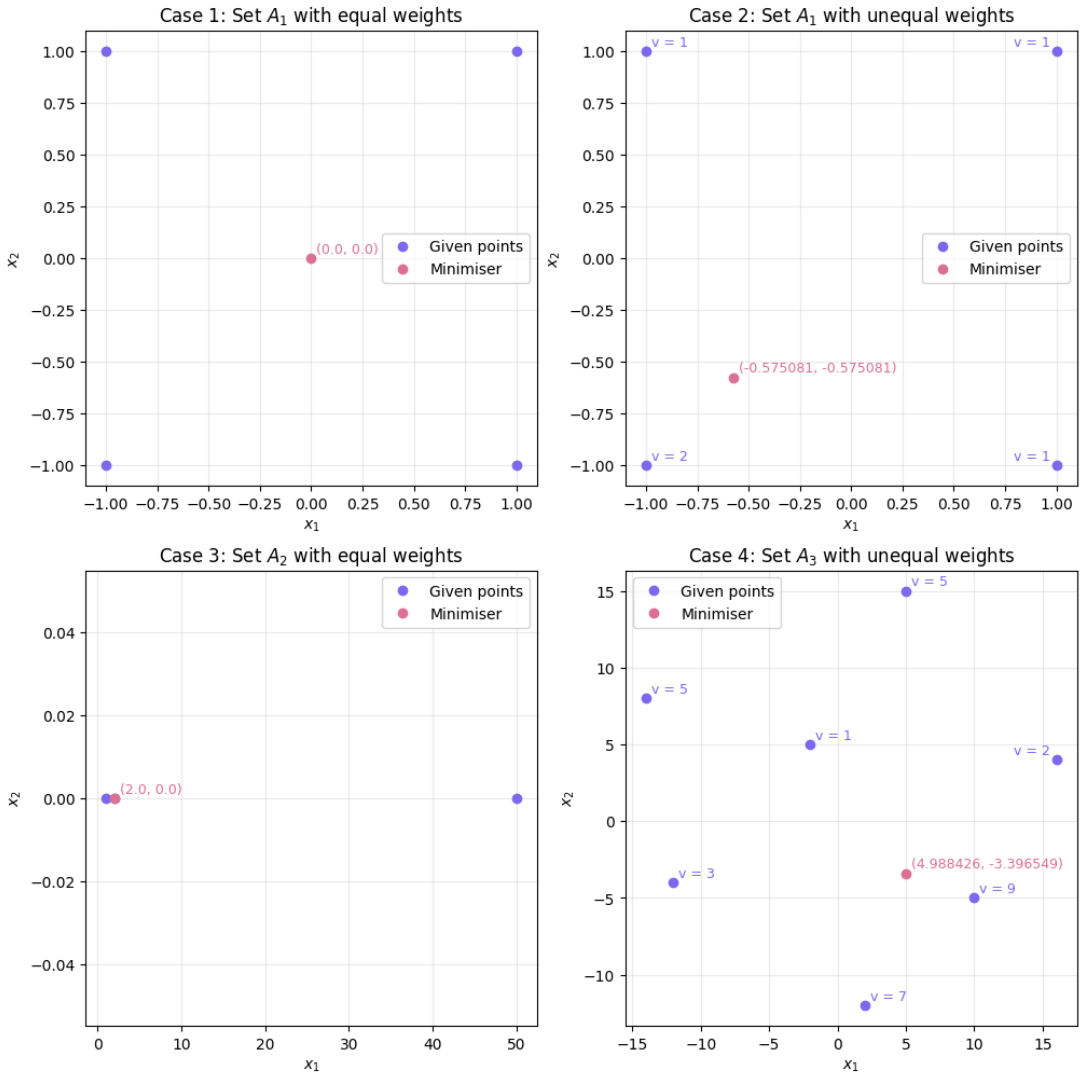


Figure 3: Plots for four different cases of given points A with weights v, and the associated minimiser of problem (3).

In case 1 (C1), the given locations are the at the corners of the unit square. Since all the weights are equal to 1, finding the minimiser reduces to solving the Median problem (2) with Euclidean distance. As mentioned earlier, for a symmetric set, the minimiser of the Median problem with Euclidean distance and the minimiser of the Median problem with squared Euclidean distance will be the same. Thus, the problem reduces to finding the mean, and the minimiser of the case is therefore located at the origin. The plot of the case in figure 3 verifies that our implementation of the Weiszfeld algorithm gives the right minimiser.

Set A_2, which is used in case 3 (C3) has been discussed earlier. Since all weights are equal to 1, the results from the previous discussion will still hold, i.e. the minimiser will be located at $(2, 0)$. The plot of the case in figure 3 verifies that the algorithm gives the right minimiser also for this case.

For both these cases, the locations in the set A are all weighted the same, and they all affect the location of the minimiser by "pulling" it towards themselves with equal weight. However, when the weights are unequal, how much "pull" each location a^i performs depends on the value of the corresponding weight. Smaller values of the weight v^i mean

less "pull", while greater values mean more "pull". In the two rightmost plots in figure 3, the weights are unequal. We observe that for case 2 (C2), where one location has double the weight of the others, the minimiser is located a lot closer to this point than in case 1 (C1), which uses the same set A. Case 4 (C4) is a more complex case than the first three, but we can observe that the minimiser is "pulled" close to the points with higher weights $v = 9$ and $v = 7$.

Gradient Descent with backtracking has also been implemented, and the implementation can be found in the Jupyter notebook. The main iteration scheme for this algorithm is

$$x_{k+1} = x_k - \alpha_k \nabla f, \tag{6}$$

which is a line search with direction $p_k = -\nabla f$. The parameter α_k is chosen with the Armijo condition

$$f(x_k + \alpha_k p_k) \geq f(x_k) + c\alpha_k \nabla f^T p_k, \quad c \in (0, 1), \tag{7}$$

where $c = 2/3$ has been used in the implementation. Using table 1, one can observe that the two algorithms perform differently for different cases. Both algorithms have been implemented with the termination criterion earlier derived (5), as both can be looked at as descent algorithms [4].

Case	Start	Weiszfeld iterations	GD iterations	Weiszfeld operations	GD operations
1	Mean	0	0	0	0
2	Mean	23	9	23	10
3	Mean	0	0	0	0
4	Mean	14	8	14	8
1	(2,2)	9	12	9	21

Table 1: The table shows number of iterations and operations for both Weiszfeld and Gradient Descent (GD) for the 4 different cases of sets A and v. For all except the last row, the mean is used as the starting point, while in the last row the algorithms uses point $(2, 2)$ as the starting point.

Case 1 (C1) is solved easily by both algorithms, as our starting point is the mean, which is equal to the solution. Thus neither of the algorithms need to start the iteration scheme, because the stopping criterion (5) is already met. This can be seen in Table 1, where number of iterations and operations for both algorithms are 0. For a different starting point further away from the solution, $x = (2, 2)$, the Weiszfeld algorithm performs better. In this case, even though our starting point is not as directly affected by the symmetry of the location points, they still contribute to the fast convergence of the Weiszfeld algorithm. In one iteration, the new iteration point becomes approximately $(0.30, 0.30)$, on the exact diagonal towards the centre. The opposite points thus still "balance" the descent directions, and we get big jumps directly towards the minimiser. Although the gradient descent also moves towards the centre, it might not do it as fast, as it needs to adjust adjust step length, which gives it more operations, as shown in Table 1.

Table 1 shows that Gradient descent finds the minimiser for case 2 (C2) and case 4 (C4) in less operations than Weiszfeld. This is because the minimiser is located near the

given location with highest weight, and the denominator from the main iteration scheme (4) will thus become big, as $d_2(a^i, x)$ becomes small. This can result in small step sizes and a slower convergence to the minimiser, as the location of one point dominates the whole fraction.

Both algorithms find the minimiser in case 3 (C3) before starting the iteration scheme, as it is one of the given points and the condition in Theorem 2 will hold. This is verified in Table 1.

In general, the two algorithms perform differently for different problems. For symmetric problems, the Weiszfeld algorithm shows faster convergence to the minimiser, as it favours symmetric problems. For problems where the weights are more unevenly distributed, the Gradient Descent proved to be more effective. One would therefore need to evaluate the given locations and weights before deciding which algorithm to use.

Conclusion

In this report we have studied a number of different location problems. Some of the problems have an analytical solution which can be easily derived using necessary optimality conditions. Other problems, namely the Median problem with Euclidean distance proved harder to solve analytically. In this case, an approximation of the minimum can be found using the Weiszfeld algorithm and Gradient Descent. Out of these two, the Weiszfeld algorithm proved better for symmetric problems, while Gradient Descent was more effective when the minimiser is located near a given location.

References

[1] Elizabeth Köbis, *Project Description for Optimization 1: Location Analysis*, chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://wiki.math.ntnu.no/_media/tma4180/2026v/project-2026.pdf

[2] Marcus Grasmair, *Theory of unconstrained optimisation*, chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.ntnu.no/emner/TMA4180/2025v/notes/UnconstrainedOptimisation_Theory.pdf

[3] S. Boyd, J. Duchi, M. Pilanci, and L. Vandenberghe, *Subgradients, Notes for EE364b, Stanford University, Spring 2021-22*, chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://web.stanford.edu/class/ee364b/lectures/subgradients_notes.pdf

[4] J. B. Rosen, G.L. Xue, *On the Convergence of Miehle's Algorithm for the Euclidean Multifacility Location Problem, University of Minnesota, 01.02.1992*, https://pubsonline.informs.org/doi/10.1287/opre.40.1.188

[5] Zhao-Rong Lai et al., *A De-singularity Subgradient Approach for the Extended Weber Location Problem, Jinan University et al. 2024*, chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.ijcai.org/proceedings/2024/0483.pdf